NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

2026-2027 (Semester 1) **MA1513** **Lecture Problem Set 6**

Instructions

- To be discussed during lecture in week 6 (Sep 14 and 15, 2026).
- Note that you do not need to submit the answers to this problem set.
- To prepare for the problem set, students should view the "Week 6 key video" before the lecture.
- Students who would like to try out the problems in advance may refer to the hints given at the end of the problem set.

1. For each of the following SDE, find a fundamental set of solutions and write down a general solution for the system.

   (a) $\begin{cases} y_1' = y_2 \\ y_2' = -y_1 \end{cases}$ (b) $\begin{cases} y_1' = -4y_1 \\ y_2' = -4y_2 \end{cases}$ (c) $\begin{cases} y_1' = 2y_1 - 2y_2 \\ y_2' = 2y_1 + 2y_2 \end{cases}$

   (d) $\begin{cases} y_1' = -y_1 + y_2 \\ y_2' = -4y_1 + 3y_2 \end{cases}$ (e) $\begin{cases} y_1' = 3y_1 + 2y_2 \\ y_2' = -8y_1 - 5y_2 \end{cases}$

2. For each of the following SDE, (i) determine the type of equilibrium point and its stability; (ii) sketch a few trajectories in the phase portrait, indicating the directions; (iii) write down the general solution.

   (a) $\boldsymbol{y}' = \begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix} \boldsymbol{y}$; (b) $\boldsymbol{y}' = \begin{pmatrix} 6 & 9 \\ 1 & 6 \end{pmatrix} \boldsymbol{y}$;

   (c) $\boldsymbol{y}' = \begin{pmatrix} 0 & 2 \\ -9 & 0 \end{pmatrix} \boldsymbol{y}$; (d) $\boldsymbol{y}' = \begin{pmatrix} -4 & 0 \\ 0 & -3 \end{pmatrix} \boldsymbol{y}$

   (e) $\boldsymbol{y}' = \begin{pmatrix} 0 & 2 \\ -5 & -2 \end{pmatrix} \boldsymbol{y}$; (f) $\boldsymbol{y}' = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \boldsymbol{y}$.

3. Given a mass-spring system that keeps oscillating without coming to a halt. Suppose the spring constant is 4 Newton/metre and the mass of the ball is 1 kg.

   (a) Express the oscillation of the system as an SDE.

   (b) Describe the equilibrium point and the stability of this system.

   (c) Find the solution of the system if the starting position of the ball is 1 metre below the rest position with an initial velocity of 1/2 metre per second. (The downward displacement and velocity are taken to be positive values.)

   (d) If we now introduce damping to the system, find the possible value(s) of the damping constant c for the ball to come to rest as quickly as possible without oscillation.

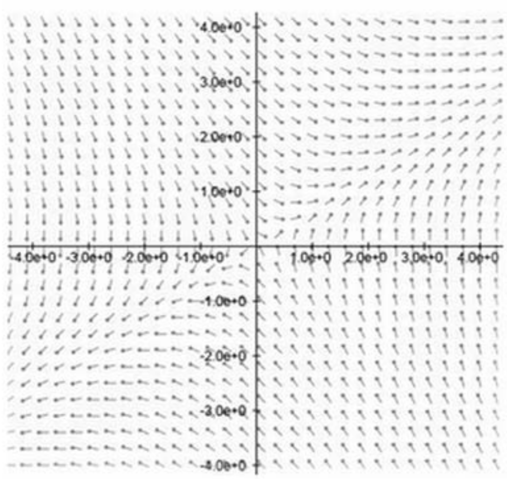
4. Without the help of any phase plane plotter app or finding the eigenvalues and eigenvectors,

   (i) match each of the following systems with the right direction field among the diagrams below;

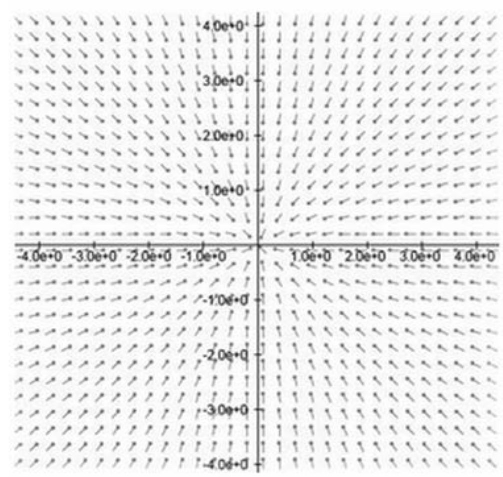
   (ii) identify the type of equilibrium point and stability associated to each diagram.

$$\text{(a)} \begin{cases} y_1' &= y_1 \\ y_2' &= y_2 \end{cases}, \quad \text{(b)} \begin{cases} y_1' &= -y_1 \\ y_2' &= -y_2 \end{cases}$$

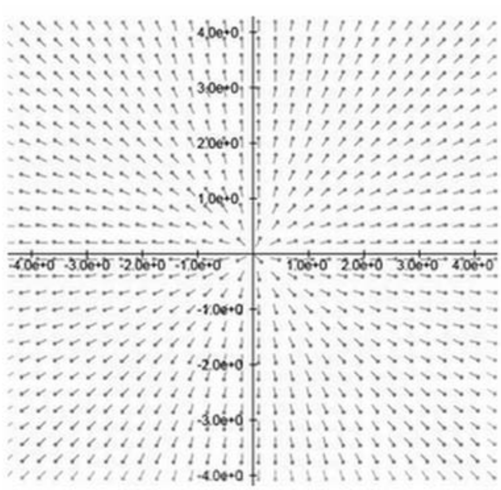
$$\text{(c)}\ \boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \boldsymbol{y}, \quad \text{(d)}\ \boldsymbol{y}' = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \boldsymbol{y}$$



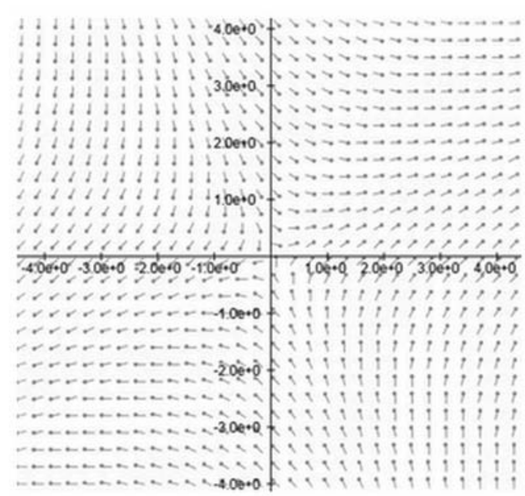
Direction field A



Direction field B



Direction field C



Direction field D

Hints and selected answers

Note: Eigenvectors of a given matrix are not unique. For example, if $\boldsymbol{v}$ is an eigenvector corresponding to a certain eigenvalue λ, then any scalar multiple $k\boldsymbol{v}$ will also be an eigenvector for λ.

The choice of your eigenvectors will affect how your fundamental set of solutions and general solution of the SDE look. But it will not affect the particular solution of an initial value problem on SDE.

1. Find the eigenvalues and eigenvectors for the underlying matrix in each case to form a fundamental set of solutions for the system.
   The general solution in each case are given below:
   (a) $y_1 = c_1 \sin t - c_2 \cos t, \quad y_2 = c_1 \cos t + c_2 \sin t$ (using eigenvectors $\begin{pmatrix} \mp i \\ 1 \end{pmatrix}$)

   (b) $y_1 = c_1 e^{-4t}, \quad y_2 = c_2 e^{-4t}$ (using eigenvectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$)

   (c) $y_1 = -c_1 e^{2t} \sin 2t + c_2 e^{2t} \cos 2t, \quad y_2 = c_1 e^{2t} \cos 2t + c_2 e^{2t} \sin 2t$ (using eigenvectors $\begin{pmatrix} \pm i \\ 1 \end{pmatrix}$)

   (d) $y_1 = \frac{1}{2} c_1 e^t + c_2 e^t (\frac{t}{2} - \frac{1}{4}), \quad y_2 = c_1 e^t + c_2 e^t t$ (using eigenvectors $\begin{pmatrix} 1/2 \\ 1 \end{pmatrix}$ with generalised eigenvector $\begin{pmatrix} -1/4 \\ 0 \end{pmatrix}$)

   (e) Use eigenvectors $\begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$ with generalised eigenvector $\begin{pmatrix} -1/8 \\ 0 \end{pmatrix}$.

2. (a) $\boldsymbol{y} = c_1 e^{-2t} \begin{pmatrix} -1/3 \\ 1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$;

   (b) $\boldsymbol{y} = c_1 e^{3t} \begin{pmatrix} -3 \\ 1 \end{pmatrix} + c_2 e^{9t} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$;

   (c) $\boldsymbol{y} = c_1 \begin{pmatrix} \sqrt{2}/3 \sin 3\sqrt{2}t \\ \cos 3\sqrt{2}t \end{pmatrix} + c_2 \begin{pmatrix} -\sqrt{2}/3 \cos 3\sqrt{2}t \\ \sin 3\sqrt{2}t \end{pmatrix}$ (using eigenvectors $\begin{pmatrix} \mp\sqrt{2}i/3 \\ 1 \end{pmatrix}$);

   (d) $\boldsymbol{y} = c_1 e^{-4t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$;

   (e) $\boldsymbol{y} = c_1 e^{-t} \begin{pmatrix} -1/5 \cos 3t + 3/5 \sin 3t \\ \cos 3t \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -1/5 \sin 3t - 3/5 \cos 3t \\ \sin 3t \end{pmatrix}$ (using eigenvectors $\begin{pmatrix} (-1 \mp 3i)/5 \\ 1 \end{pmatrix}$);

   (f) $\boldsymbol{y} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

3. Refer to Chapter 4 notes page 21-23.
   (a) $\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -k & 0 \end{pmatrix} \boldsymbol{y}$ (b) Centre; neutrally stable

   (c) $\boldsymbol{y} = \begin{pmatrix} \frac{1}{2\sqrt{k}} \sin \sqrt{k}t + \cos \sqrt{k}t \\ \frac{1}{2} \cos \sqrt{k}t - \sqrt{k} \sin \sqrt{k}t \end{pmatrix}$ (d) $c = 2\sqrt{k}$.

4. (i) Test with some appropriate points on the phase plane and pre-multiply the coordinates of the point with the underlying matrix of the system to see the corresponding tangent vector. Refer to Chapter 4 notes page 12-13.
(ii) Trace a few trajectories following the flow of the direction fields to identify the type of equilibrium points.